

1.0 GENERAL REQUIREMENTS

Lube oil system to be designed as per the required specification to meet operational requirement keeping Noise, Vibration and other statutory requirements.

Lube oil systems (fully assembled skids with all instrumentation, electrical components and piping in line with API 614 5TH Edition)

OIL TYPE: Servo prime ISO VG-46

System should deliver following parameters at output terminals:

Parameters at Turbine skid terminal;

Lube Oil – **50 LPM, 1.8 – 2.0 Kg/cm² g.**

Note: Above Parameters are actual requirements, vendor to consider the margin as per **API-614 5TH edition (Chapter 1 & 3)** & design the system accordingly.

However, the vendor must fulfill its obligations by considering API 614 5th Edition along with the specifications.

2.0 SCOPE

This specification covers the minimum requirements for lubrication systems and auxiliaries.

3.0 TECHNICAL PARAMETERS.

Main oil tank:

- Capacity: As per API 614
- Material: SS 316

Main oil pump (Not in LOS vendor scope):

- Type: Screw pump.
- Operating speed: 1500 rpm.
- Capacity: 50 Liters/minute + margin.
- Driver: Gear box low speed shaft

Auxiliary oil pump:

- Type: Screw pump
- Operating speed: 1500 rpm
- Capacity: 50 Liters/minute + margin.
- Driver: AC motor (TEFC / IE3)

Emergency oil pump:

- Type: Screw pump
- Capacity: 30% of MOP Capacity or 20 LPM (Whichever is maximum).
- Driver: UPS AC motor (TEFC / IE3)

Lube Oil cooler:

- Design code: ASME Sec VIII Div. 1
- Oil cooler Type: 2x100%, TEMA-C type Shell and tube duplex removable-bundle coolers.
- Cooler tube type & Size: Seamless tube removable-bundle type, Tube OD 3/4" & Thick. 16 BWG
- CW Supply Temperature: 33° C
- CW Return Temperature: 45° C(max), however minimum of 4-5 °c Δt to be considered,
- CW Inlet Pressure: 5.0/6.5 Kg/cm² g (Nor/Max)
- Tube Side design pressure & temperature shall be 10.2 kg/cm² g & 100 °c.
- Oil inlet temperature: 70°C

- Oil outlet temperature: 45°C
- Oil flow Design Capacity: 50 Liters/minute (Normal) + margin.
- Cooler shall be 30% over designed considering pump max. flow.
- Pressure drop at tube side shall be limited to 0.6 kg/cm²(max).
- U-bend tubes are not permitted.
- Fouling factor 0.0004 hr m² EC/kcal (water side) & 0.0002 hr m² EC/kcal (Oil side).

Note:

- a. Low pressure (LP –Tube Side) side shall be preferably specified with a design pressure at least equal to 10/13 of high pressure (HP – Oil side) side design pressure.
- b. To prevent the oil from being contaminated if the heat exchanger fails the oil-side operating pressure shall be higher than the water-side operating pressure.
- c. CW water line piping ~8m length with fittings, instruments with hook ups (TG & PG) , safety valve, Flow glass, valves, drain valves, back flush line & sampling point to be included in scope & shall be part of the LOS skid. Each utility such as cooling water supply and return lines and others as specified shall be manifolded to a common connection.
- d. All vent & drains of the cooler shall be connected back to the tank with orifice plate, valves & flow glass.
- e. TEMA datasheet for the cooler to be submitted.
- f. Heat exchanger oil drains shall be manifolded, but not together with filter clean side drains.

Lube oil filter:

Filter Type: Duplex Oil filter (1 working + 1 standby)

Oil flow Capacity: 50 Liters/minute + Margin

Oil filtration Capacity: 10 - 15 microns.

Pressure drop @ Clean/dirty condition: 0.35/0.6 Kg/cm².

Note:

- a. Filters with replaceable elements or cartridges shall be provided.
- b. All connections on filter housings shall be flanged
- c. Filter housings shall be equipped with valved vents and clean- and dirty-side valved drain connections.
- d. Vents & drains (Clean side oil) of the filter shall be connected back to the tank with valves, orifice plate & flow glass.
- e. Drain connections (Dirty side oil) shall be blind-flanged.
- f. Sample point shall be provided with valve.

Pressure control valve:

Set pressure: As per suitability

MOC- Body & Bonnet – As per PMS (If not specified same shall be as per pipe MOC).

Trim- As per PMS (If not specified same shall be SS-316 minimum).

Note:

- a. PCV shall include upstream, downstream, bypass & drain valves with blind flanged end.
- b. Outlet line of the PCV shall include flow glass before connecting back to tank.

Temperature control valve:

Set pressure: As per suitability

MOC- Body & Bonnet – As per PMS (If not specified the same shall be similar to pipe MOC)

Trim- As per PMS (If not specified same shall be SS-316 minimum).

Note:

- a. Temperature control valve shall be an internal thermostat-operated three-port valve.

- b. Temperature control valve shall be provided with a manual override that permits operation independent of temperature conditions.
- c. TCV shall include upstream, downstream, bypass & drain valves with blind flanged end.

Pressure safety valve:

Set pressure: As per suitability

MOC- Body & Bonnet – As per PMS (If not specified the same shall be similar to pipe MOC).

Trim- As per PMS (If not specified same shall be SS-316 minimum).

Note:

- a. Isolation valve shall be provided at the upstream of the PSV.
- b. Outlet line of the PSV shall include flow glass before connecting back to tank.
- c. Minimum pressure rating of the valve shall be 300#.

Overhead tank:

Only provision on main oil tank shall be provided.

4.0 MATERIALS:

a. Pumps:

- Casing : Fabricated Steel
- Gear / Rotor : 16CrMn15

b. Lube Oil cooler:

- Cooler shell, channels, covers: SA 516 Gr.70 / A 106 Gr B.
- Cooler tube material: Inhibited admiralty brass. (ASTM B111)
- Cooler Baffle/ Tube sheet material/Tie Rod: Shall be compatible with the tube material.
- Cooling water line – As per PMS (If not specified the same shall be similar to pipe MOC).

c. Lube oil filter:

- Filter Body: SS316.
- Filter Element: Glass fiber / Micro-felt.

d. Main oil reservoir: SS 316

e. Oil piping & fittings: SS 316

f. Valves, Strainers & other Accessories: As per PMS (If not specified the same shall be similar to pipe MOC).

g. Gaskets: As per PMS (If not specified the same shall be SS316 (ID Ring) + Grafoil + SS316 (OD Ring)).

5.0 BRIEF TECHNICAL REQUIREMENT OF ELECTRICAL ITEMS

a. Motors:

- Safe area weather proof protection for MV/LV motors shall be standard requirement.
- Hazardous area Zone – Safe area.
- All motors & heaters, LPBS are certified with project specific requirement.
- LPBS & Motor LCS shall be as per safe area applications.

b. Instruments :

- All instruments need to consider for hazardous area zone-1 IIC, T3.
- All instruments are with HART-7 protocol (analog 4-20mA).
- All instruments & electrical items need to designed considering (+) 1.1 °C.

- c. Lubrication oil online moisture and particle analyser.

6.0 BASIC SYSTEM FEATURES

Oil tank, oil cooler (including CW piping), oil pump and oil filter to be assembled in single skid only. Turbine is kept at +2.00 meter floor and Lube oil system placed at +0.00 meter floor.

OPERATIONAL PHILOSOPHY:

Case 1: Lube oil supply to bearings of Main Steam turbine, Gearbox is to be considered as normal condition.

Case 2: With the unavailability of the steam (Due to shut down or trip), bearing oil supply to turbine, Gearbox will be functional for at least 30 mins of duration and will be by-passed to the lube oil reservoir.

7.0 OPERATION, MAINTENANCE, TESTING, INSPECTION, SHIPPING & DISPATCH, INSTALLATION, STORAGE ETC.

For cooling water parameters, Location refer project specific documents (Utility data / Basic Engineering design data).

For other requirements refer to project specific documents & standards attached.

6.0 REQUIRED DOCUMENTS.

S.No	Type of Document	Number of Copies	Status	Time schedule
1	GA		For Approval	With offer
2	P&IDs, BOM		For Approval	With offer
3	Lube oil tank & Run down tank sizing details		Documentation	Within 3-4 weeks of PO
4	Data sheets as per customer format (Pump, Motor, Cooler, Filter, Valves & Accumulator, PCV, PSV, TCV, Oil Heater)		Documentation	Within 3-4 weeks of PO
5	Instrument list & datasheets as per customer format		Documentation	Within 3-4 weeks of PO
6	QAP (Preliminary)		For Approval	With offer
7	QAP with full details		For Approval	Within 3-4 weeks of PO
8	3D Models (.STP format)		For Approval	With offer
9	Mechanical calculations for Filter, Accumulator & Cooler.		Documentation	Within 3-4 weeks of PO
10	Lubricant list		For Approval	With offer
11	Tags list for all the components for identification.		For Approval	Within 5-7 weeks of PO
12	Project specific spare parts list & document for spare part interchangeability (SPIR)		Documentation	With offer

13	Equipment Record Cards or ERCs" which are basically As Built data sheets. This is a mandatory pre requisite for the Factory Acceptance Test (FAT) which is the Mechanical Run Test (MRT) for Steam Turbine and Lube Oil System		Documentation	Before Dispatch
14	Wiring diagram with JB GAD		For Approval	Within 3-4 weeks of PO
15	LGB drawing with hook up diagram		For Approval	Within 3-4 weeks of PO
16	Piping support details		For Approval	Within 3-4 weeks of PO
17	Lifting sketch & arrangement		For Approval	Within 3-4 weeks of PO
18	Packing procedure		For Approval	Within 3-4 weeks of PO
19	Painting procedure		For Approval	Within 3-4 weeks of PO
20	Preservation procedure			Within 3-4 weeks of PO
22	Electrical load list		For Approval	Within 3-4 weeks of PO
21	Weld Book		For Approval	Within 3-4 weeks of PO

IMPORTANT NOTE:

1. Other relevant documents related above standards will be referred from the project specific documents.
2. LOS skid shall be suitable for (+) 1.1 °C MDMT.
3. LOS skid will be kept in open atmosphere (outdoor application) hence same to be designed accordingly.
4. Seismic & wind load details shall be provided by the vendor during detail engineering.
5. We will not review the vendor makes in the offer. Vendor to adhere to the project specific AVL provided for the bought out items.
6. Provided LOS specification along with the project specifications & vendor P ID (as per the min. recommendation of API 614) are the final if any contradiction with the respect to the scope.
7. Provision of lube oil return nozzle shall be made at the side wise of the main oil tank.
8. Junction boxes, LCS shall be considered as SS316L minimum as per the project requirement.
9. All transmitter and pressure gauges drain need to connect to the single common header.